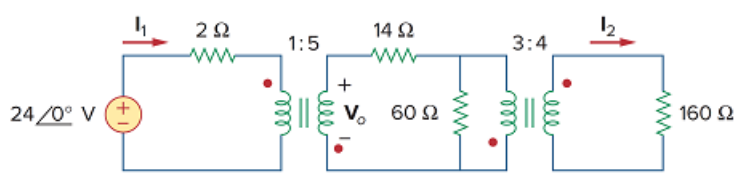


Problem

For the circuit in Fig. 13.126, find I_1 , I_2 , and V_o .

Figure 13.126



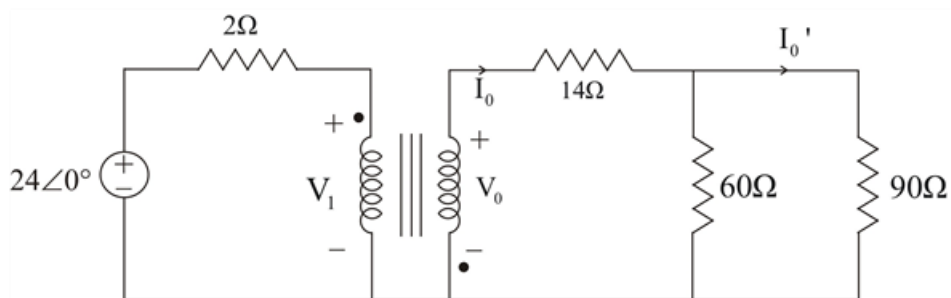
Step-by-step solution

Step 1 of 3

We reflect the 160 – ohm load the middle circuit.

$$Z_R = \frac{Z_L}{n^2} = \frac{160}{\left(\frac{4}{3}\right)^2} = 90 \text{ Ohms} \quad \text{Where } n = \left(\frac{4}{3}\right)$$

Step 2 of 3



Step 3 of 3

$$14 + 60 \parallel 90 = 14 + 36 = 50 \text{ Ohms}$$

We reflect this to the primary side,

$$Z_R' = \frac{Z_L'}{(n')^2} = \frac{50}{5^2} = 2 \text{ Ohms}, \quad n' = 5$$

$$I_1 = \frac{24}{(2+2)} = 6 \text{ A}$$

$$24 = 2I_1 + V_1$$

Or,

$$V_1 = 24 - 2I_1 = 12 \text{ V}$$

$$V_0 = -nV_1 = -60V_1,$$

$$I_0 = \frac{-I_1}{n_1} = \frac{-6}{5} = 1.2$$

$$I_0' = \left[\frac{60}{(60+90)} \right] I_0 = -0.48 \text{ A}$$

$$I_2 = \frac{-I_0'}{n} = \frac{0.48}{\left(\frac{4}{3}\right)} = 0.36 \text{ A}.$$